ROLLING CONTACT BEARINGS

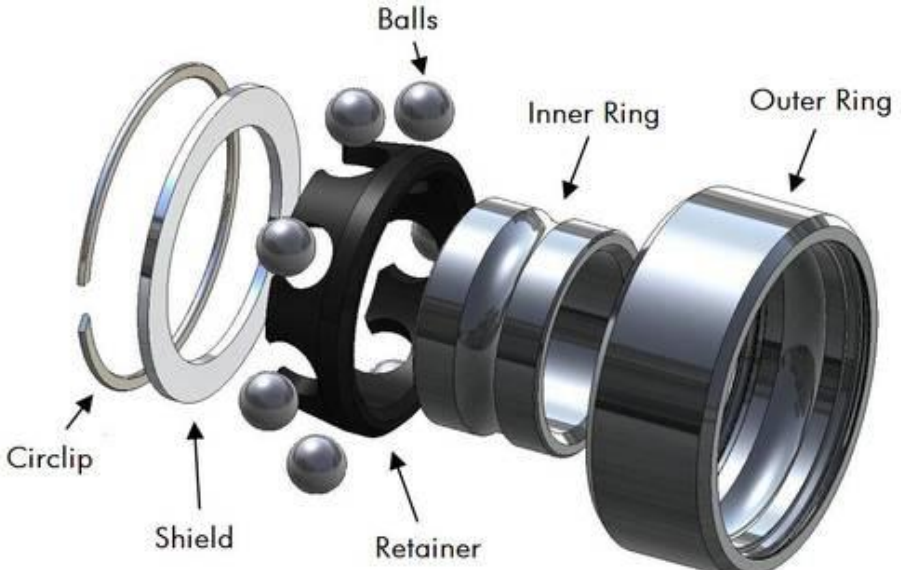
In this bearing, contact between the bearing elements is rolling instead of sliding as in the plain bearings.

In rolling-contact bearings, the shaft and outer members are separated by balls or rollers, and thus rolling friction is substituted for sliding friction.

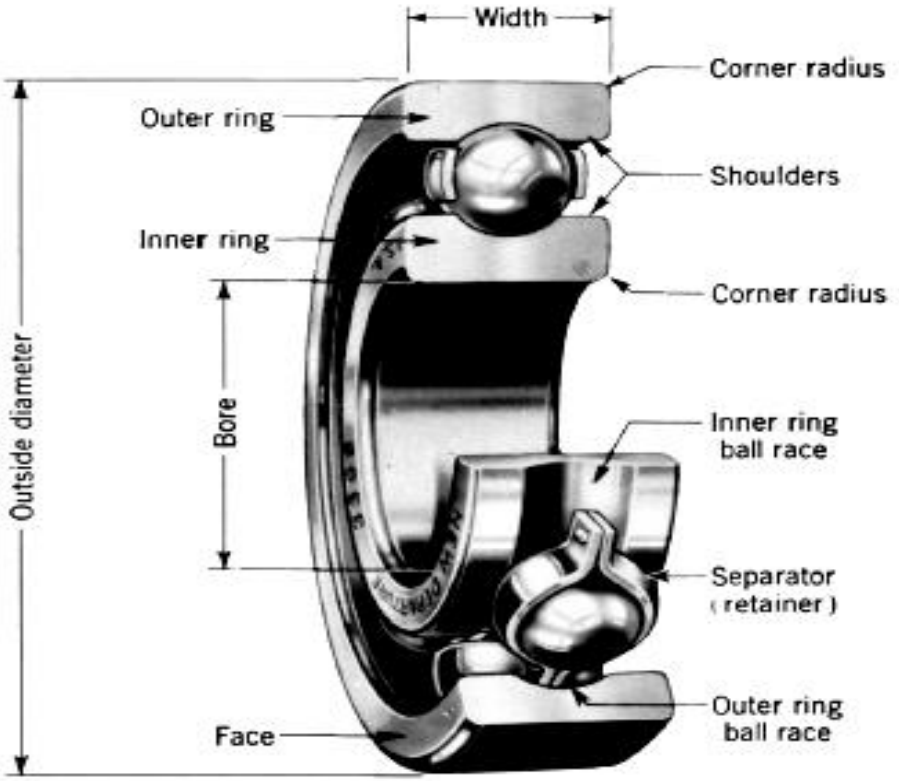
Since the rolling friction is very less as compared to sliding friction these bearings are also called “Anti-Friction Bearings”

The contact areas are small and the stresses are high, the loaded parts of rolling-element bearings are normally made of hard, high-strength materials, superior to those of the shaft and outer member (These parts include inner and outer rings and the balls or rollers)

A retainer or separator, which keeps the balls or rollers evenly spaced and separated.



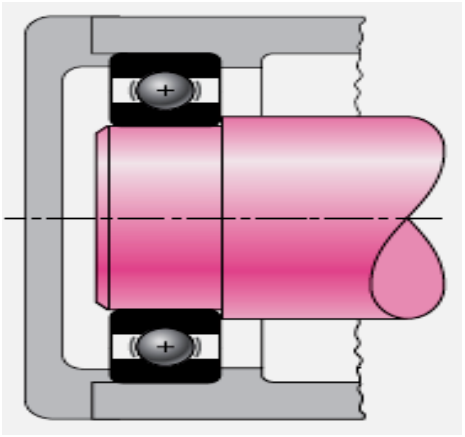
http://www.grw.de/en/products.html



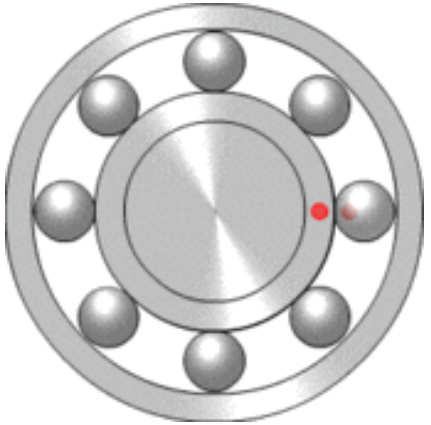
Schematic of radial ball bearing



www.sabearings.com



Assembled position of ball bearing



http://simple.wikipedia.org/wiki/Bearing_%28mechanical%29

Two sets of the widest and heaviest spherical roller bearings in LYCs history for Metallurgical & Mining Industry were off line in LYC wind turbine bearing manufacturing factory.

The weight of each bearing is up to 5.5 tons and width is 630mm and the weight of each roller is 30kg.



Classifications



Ball bearings: Rolling element is a spherical ball

Roller bearings: Rolling element is roller which might be cylindrical, conical, spherical or concave

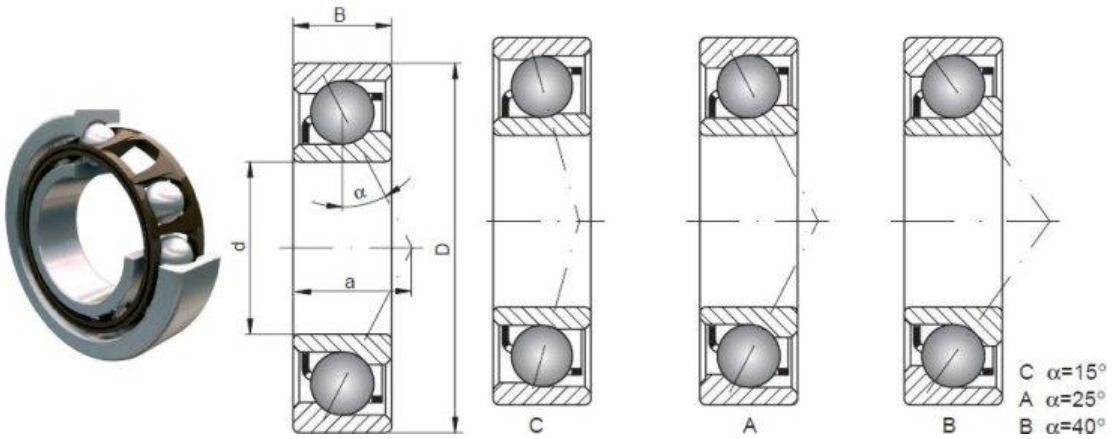


1. Radial Ball Bearings
   a) Deep groove ball bearing

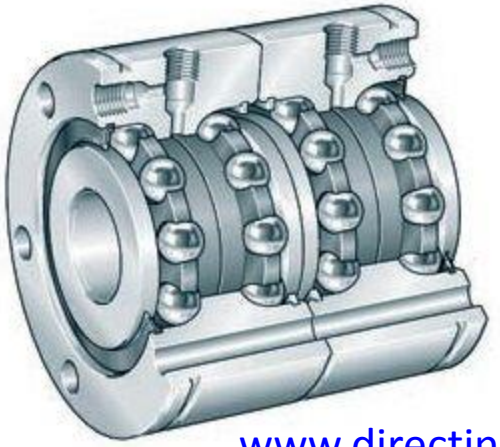
   b) Self aligning ball bearings



2. Angular Ball Bearings



3. Thrust ball bearings

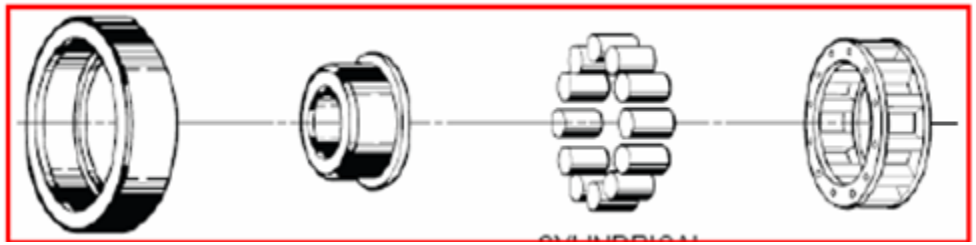


catalog.ubx-usa.com

www.directindustry.com

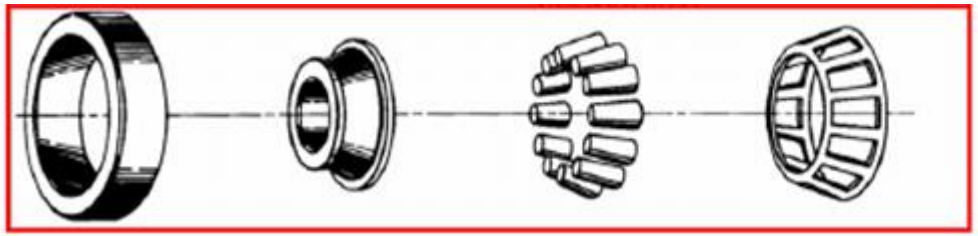
Roller bearings

1. Cylindrical roller bearings: Very high radial load not thrust
2. Tapered roller bearings: Heavy radial and axial loads
3. Spherical roller bearings: Self aligning, heavy radial load and less thrust
4. Needle bearings : in automobile components such as rocker arm pivots pumps, compressors, and transmissions.



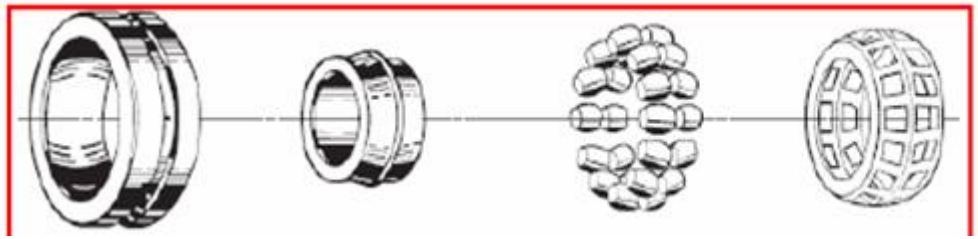
- Cylindrical roller





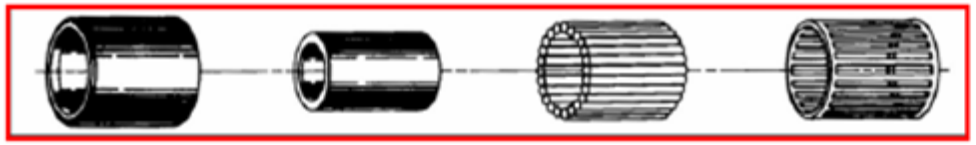
-Tapered roller





- Spherical roller



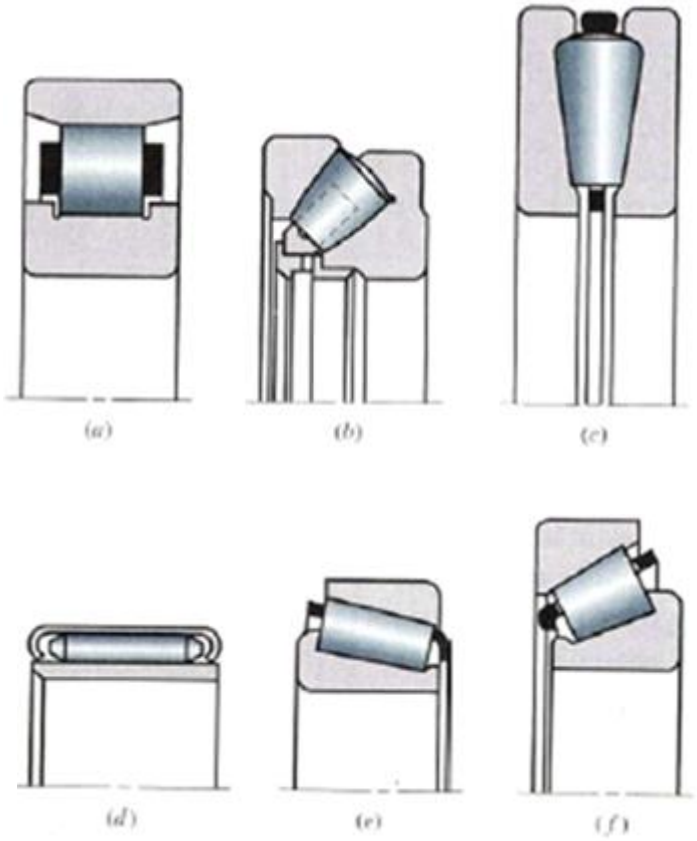


- Needle roller



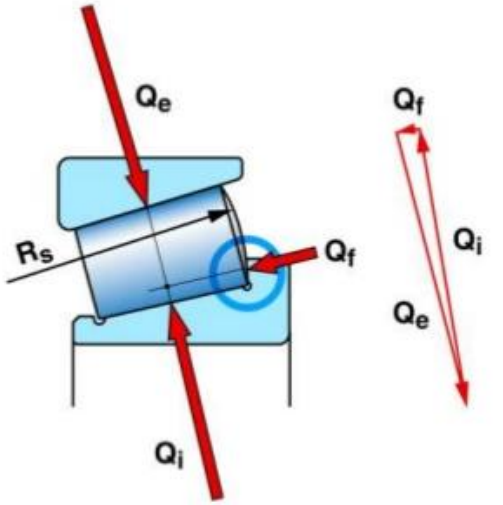
Cont…

Roller bearings



- Roller bearing are usually used when shock and impact are present, or when large bearing are needed.
- Tapered roller bearing can carry a large axial load. The magnitude depends on the angularity of the rollers. The radial load will also produce a thrust component.
- Roller bearing in general can be applied only where the angular misalignment caused by shaft deflection is very small
- Spherical roller bearing has excellent load capacity and carry a thrust component in either direction. high misalignment

Roller Forces



Q_e = Outer ring/ roller contact force

Q_i = Inner ring/ roller contact force

Q_f = Roller/ flange contact force

Effects of Axial and Radial Load



Q_R = External radial load

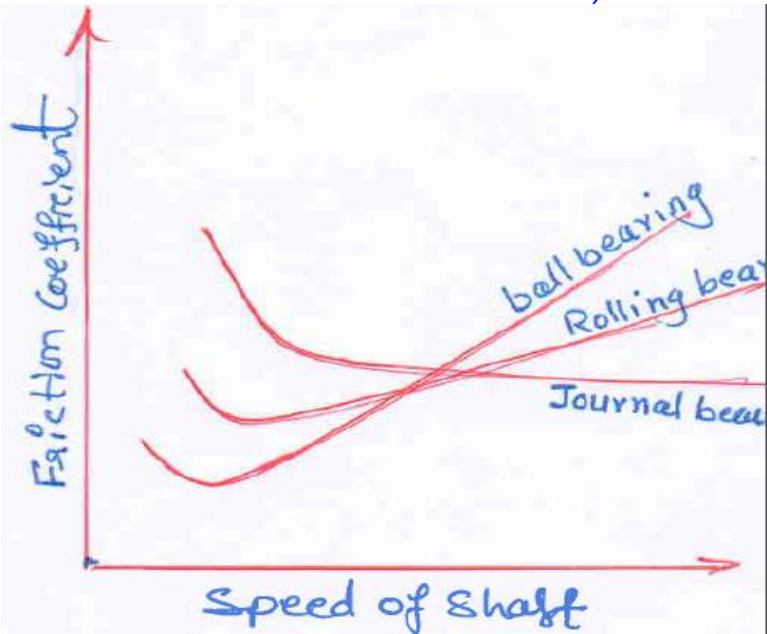
Q_A = Internal induced axial load

Advantages of Rolling bearings compared to Journal bearing

- Low starting friction torque than the running friction torque.
- Good low-temperature starting
- Low maintenance, compactness and readily replaceable
- Can be pre-loaded causing a shaft to be stiffer.
- Ease of lubrication, can seal lubricant and lifetime-lubricate
- No self exited instabilities
- Less axial space for a comparable shaft diameter
- Shaft alignment is more accurate than in the sliding bearings.
- It can carry both radial as well as thrust loads
- Preferable in precision machinery (available from few mm to 400mm)

Disadvantages of Rolling bearings

- Finite life due to eventual failure by fatigue
- Lesser capacity to withstand shock
- Initial cost is high
- Greater diametral space
- Noise and coefficient of friction is more at higher speeds.



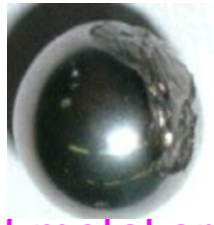
Bearing Failure

- The bearings which are carefully installed, properly lubricated, kept free of foreign matter and not overloaded will still fail due to fatigue.

- Rolling elements revolve within the raceways hence, they are subjected to repeating contact stress.

- Even for operation under controlled conditions fatigue tests of bearings which are identical show different lives.

- The rolling bearing has a limited life, limited by number of repetitions of stress to cause fatigue failure.
  Failure of lubricant may also cause the failure of bearings.



- Since pure rolling does not occur in the bearing, small particles of metal are removed, which is called **spalling failure**

- Thus it is difficult to predict the life of an individual bearing, and the only logical approach is from statistics and probabilities

Bearing Life

Life: defined as the number of revolutions that the bearing runs before the **first evidence of fatigue** appears on any of the moving elements.

Since the life of single bearing cannot be predicted, the life of bearing is defined in terms of performance of a group of bearings.

Bearings are rated depending upon

a) Average life or Median life of a group of bearings
b) Rated life of a group of bearings

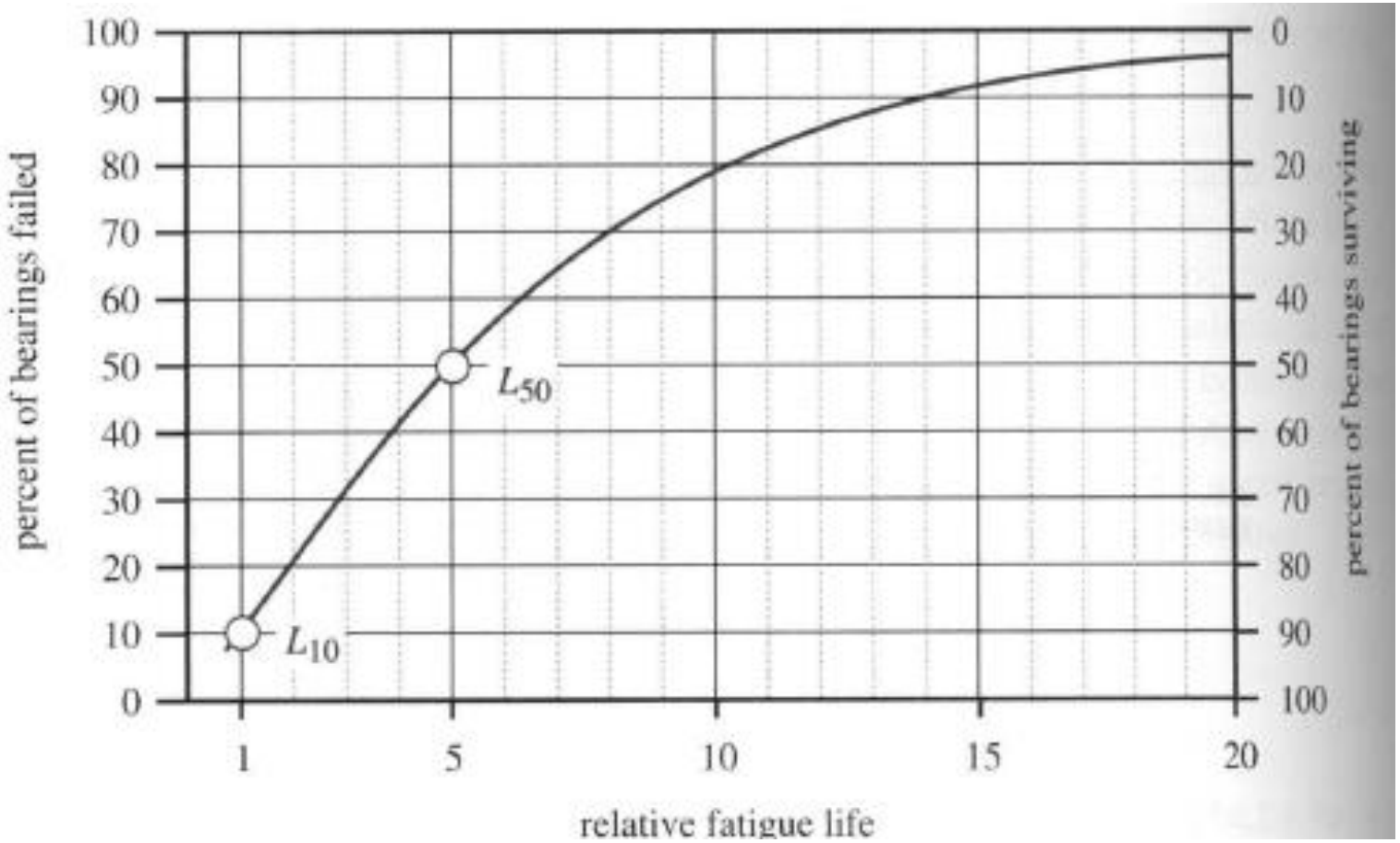
The number of millions of revolutions at which 10% of the bearings have displayed some evidence of fatigue is taken as one unit of the ordinate in Figure.

Average life or Median life of a group of particular bearings is defined as the number of revolutions (millions) that 50% of the bearing will complete or exceed at constant speed before the first evidence of fatigue appears.

Rating life of a group of identical bearings is defined as the number of revolutions that 90% of the bearing will complete or exceed before the first evidence of fatigue appears.

Average life = 5 X Rating life = $5L_{10}$

In bearing industry terminology, the number of revolutions necessary to cause failure of 10% of the bearings is known as the L_{10} life and so on. (Fig. Shows that L_{50} life is 5 times the L_{10} life .



Basic static loading: The basic static load C_o is defined as the static radial load which corresponds to a total permanent deformation of ball and race at the most heavily stressed contact of 0.0001 (permanent deformation) of the ball or roller diameter.

The magnitude of basic static load rating is given by

$$C_0 = f_0 i Z D^2 \cos\alpha \qquad \text{Eq.16.12/328}$$

where,

f_o – A constant = 3.334 for self aligning ball bearings
= 12.258 for redial contact and angular contact groove ball bearings

i – No. Rows of ball, D – Ball diameter

α - Normal angle of contact in degrees, **Z** – No. Balls per Rows,

Basic load rating /Basic load rating dynamic (C)

Basic load rating /Dynamic load rating is that constant stationary radial load which a group of apparently identical ball bearings with stationary outer ring can endure for a rating life of 1 million revolution of the inner ring.

,

The rating life in millions of revolutions:

For Ball bearings $$L = \left(\frac{C}{P}\right)^3$$

For Roller bearings $$L = \left(\frac{C}{P}\right)^{\frac{10}{3}}$$

Where, P – Equivalent Dynamic load

Equivalent Dynamic load is that constant stationary radial load which if applied to a bearing with rotating inner ring and stationary outer ring would give the same life as that which the bearing will attain under the actual condition of load and rotation.

$$P = \left(XVF_r + YF_a \right) K_s K_t$$

X and X_o : radial load factor, Y and Y_o : axial load factor

V : race rotation factor, 1.00 for inner race

1.2 for outer race

F_r and F_a : radial and axial load respectively

K_t : Temperature factor 1 for upto 100 C (Ref T16.8/339)

K_s : Shock factor or service factor, 1 for steady load,

1.25 for light shock, 1.5 for medium shock,

2 for heavy shock

Equivalent Static load

$$P_o = \left(X_o F_r + Y_o F_a \right)$$

Relationship between the life in millions of revolutions L_{10} and the life in hours L_h is given by

$$L_h = \frac{10^6}{60N}\left(\frac{C}{P}\right)^m = \frac{10^6 L_{10}}{60N}$$

N- speed in rpm

m- 3 for ball bearing
10/3 for roller bearing

Standard dimensions and designation of Ball Bearings

The Ball Bearings are made to standard dimensions of bore, outside diameter, width and fillet sizes. Standard ball bearings are available in the following series.

Extra light series or 100 series (Equivalent to dimension series 01)

Light series or 200 series (" " 02)

Medium series or 300 series (" " 03)

heavy series or 400 series (" " 04)

Bearings are designated by atleast three digit number, additional digits will indicate special features.

The last three digits indicate the series and bore size.

The last two digits multiplied by 5 yield the bore diameter in mm.

Thus, if a bearing is designated by the number 208, it is a light series bearing with bore 08 X 5 = 40mm.

The Antifriction Bearing Manufacture's Association (AFBMA) standard for the above bearing is 40 BC 02. Letter indicates the type of bearing 40 indicates the bore size and 02 indicates the series i.e., light series. (BC- Bearings and Cartridge assemblies)

Dynamic Load rating for Rolling contact Bearings under Variable Loads

The Approximate rating life for Ball bearings $L = \left(\frac{C}{P}\right)^k$

The relationship between the life in revolutions (L) and life in working hours L_H is given by

$$L = 60 N . L_H$$

Considering bearing subjected to variable loads.

Let P_1, P_2, P_3, P_4 etc., be the loads on the bearing for successive speeds n_1, n_2, n_3, n_4 etc.,

If the bearing is operated exclusively at the constant load P1, then its life is given by

$$L_1 = \left(\frac{C}{P_1}\right)^k$$

Fraction of life consumed with load P1 acting for n1 number of revolution is

$$\frac{n_1}{L_1} = n_1 \left(\frac{P_1}{C}\right)^k \times \frac{1}{10^6}$$

Similarly, fraction of life consumed with load P2 acting for n2 number of revolution is

$$\frac{n_2}{L_2} = n_2\left(\frac{P_2}{C}\right)^k \times \frac{1}{10^6}$$

Similarly, fraction of life consumed with load P3 acting for n3 number of revolution is

$$\frac{n_3}{L_3} = n_3\left(\frac{P_3}{C}\right)^k \times \frac{1}{10^6}$$

$$\text{But} \quad \frac{n_1}{L_1} + \frac{n_2}{L_2} + \frac{n_3}{L_3} + ... = 1$$

$$\text{OR} \quad n_1\left(\frac{P_1}{C}\right)^k \times \frac{1}{10^6} + n_2\left(\frac{P_2}{C}\right)^k \times \frac{1}{10^6} + n_3\left(\frac{P_3}{C}\right)^k \times \frac{1}{10^6} + ... = 1$$

$$\therefore \quad n_1\left(P_1\right)^k + n_2\left(P_2\right)^k + n_3\left(P_3\right)^k + ... = C^k \times 10^6 ..... \qquad \text{(i)}$$

If equivalent load (P) is acting for n number of revolution, then $n = \left(\frac{C}{P}\right)^k \times 10^6$

$$\therefore \quad n\left(P\right)^k = C^k \times 10^6, \qquad \text{where } n = n_1 + n_2 + n_3 + ...$$

From (i) and (ii)

$$\therefore \quad n_1(P_1)^k + n_2(P_2)^k + n_3(P_3)^k + ... = n(P)^k$$

$$\therefore \quad P = \left[\frac{n_1(P_1)^k + n_2(P_2)^k + n_3(P_3)^k + ...}{n} \right]^{1/k}$$

Substituting $n=n_1+n_2+n_3+...$, and $k=3$ for ball bearings, we have

$$\therefore \quad P = \left[\frac{n_1(P_1)^3 + n_2(P_2)^3 + n_3(P_3)^3 + ...}{n_1+n_2+n_3+....} \right]^{1/3}$$

Where, P – Equivalent Dynamic load

Reliability of a Bearing

Reliability is defined as the ratio of the number of bearings which have successfully completed L million revolutions to the total number of bearings under test.

According to Wiebull, the relation between the bearing life and the reliability is given as

$$\log_e\left(\frac{1}{R}\right)=\left(\frac{L}{a}\right)^b \qquad \text{OR} \qquad \left(\frac{L}{a}\right)=\left[\log_e\left(\frac{1}{R}\right)\right]^{1/b} \qquad \text{(i)}$$

Where L is the life of the bearing corresponding to the designed reliability R and a and b are constants whose values are

a=6.84 and b= 1.17

If L is the life of the bearing corresponding to a reliability of 90% (i.e. R_{90}), then

$$\left(\frac{L_{90}}{a}\right)=\left[\log_e\left(\frac{1}{R_{90}}\right)\right]^{1/b} \qquad \text{(ii)}$$

Dividing (i) by (ii)

$$\left(\frac{L}{L_{90}}\right)=\left[\frac{\log_e\left(1/R\right)}{\log_e\left(1/R_{90}\right)}\right]^{1/b}=8.85\left[\log_e\left(1/R\right)\right]^{1/1.17}$$